

Contestant Number: _____

Time: _____

Rank: _____



Network Administration Using Cisco (315)

STATE 2025

CONCEPT KNOWLEDGE:

Multiple Choice (50 @ 2 points each) _____ (100 points)

Test Time: 60 minutes

Multiple Choice Questions

Directions: Identify the letter of the choice that best completes the statement or answers the question.

1. During a technology upgrade, a school decided to implement a new wireless networking standard that offers high throughput across greater distances. The network must support speeds significantly faster than 54 Mb/s. Which of the following standards would best fit their requirements?
 - A. 802.11g
 - B. 802.11n
 - C. 802.11ac
 - D. 802.11b
2. How does a router determine the best path for forwarding data packets in dynamic routing?
 - A. By consulting the routing table and using routing protocols
 - B. By using MAC addresses
 - C. By learning IP addresses from incoming frames
 - D. By manually configuring static routes
3. If a network administrator wants to filter traffic based on MAC addresses at Layer 2, which feature should be used on a Cisco switch?
 - A. VLAN
 - B. Access Control Lists
 - C. Port Security
 - D. Routing Protocols
4. If a network administrator wants to monitor network traffic and analyze packet data, which network protocol should be used?
 - A. FTP
 - B. SNMP
 - C. TCP
 - D. Wireshark
5. In a network environment, what is the primary purpose of FHRP load balancing?
 - A. To optimize traffic distribution among routers
 - B. To encrypt all network traffic
 - C. To block unauthorized access
 - D. To prioritize network traffic
6. In a network environment, what is the primary role of a VLAN access list?
 - A. To control traffic between VLANs
 - B. To encrypt all network traffic
 - C. To assign IP addresses to VLANs
 - D. To manage DHCP requests
7. In a network setup, what is the primary function of VLAN routing?

- A. To enable communication between different VLANs
 - B. To encrypt all network traffic
 - C. To filter incoming traffic
 - D. To prioritize network traffic
8. In a network with multiple routers, if Router A's GigabitEthernet0/0 interface IP address is 192.168.1.1/24 and is connected to Router B's GigabitEthernet0/1 interface with IP address 192.168.1.2/24, which of the following commands will configure a static route on Router A to the IPv4 network 10.0.0.0/24 using the next-hop address of 192.168.1.2?
- A. ip route 10.0.0.0 255.255.255.0 192.168.1.2
 - B. route 10.0.0.0 255.255.255.0 192.168.1.2
 - C. ip route 10.0.0.0 255.255.255.0 GigabitEthernet0/1
 - D. route add 10.0.0.0 mask 255.255.255.0 192.168.1.2
9. In a network, if a switch receives a frame with an unknown destination MAC address, what action will the switch take?
- A. Drop the frame
 - B. Forward the frame to all ports
 - C. Send an ARP request
 - D. Broadcast the frame to all devices on the network
10. In a WAN setup, if Router A's Serial0/0/0 interface IP address is 10.1.1.1/30 and is connected to Router B's Serial0/0/0 interface with IP address 10.1.1.2/30, which command would establish an EIGRP neighbor relationship between the two routers?
- A. router eigrp 1
 - B. network 10.1.1.0 0.0.0.3
 - C. neighbor 10.1.1.2
 - D. ip eigrp 1
11. In an IPv6 network, if a router's interface is configured with address 2001:DB8:ACAD::1/64, which command should be used to enable IPv6 routing on the router?
- A. ipv6 unicast-routing
 - B. enable ipv6 routing
 - C. ipv6 routing
 - D. enable ipv6 unicast-routing
12. In dynamic routing, which process allows routers to exchange routing information with neighboring routers?
- A. Subnetting
 - B. Routing Table
 - C. Convergence
 - D. Routing Protocol

13. In IPv6, what is the purpose of the Anycast address?
- A. To communicate with the nearest device in a group
 - B. To secure the router from unauthorized access
 - C. To manage network traffic based on IP addresses
 - D. To prioritize network traffic based on specific criteria
14. In router configuration, what is the purpose of setting up NAT (Network Address Translation)?
- A. To divide a network into separate broadcast domains
 - B. To determine the best path for data transmission
 - C. To translate private IP addresses to public IP addresses
 - D. To manage network bandwidth
15. In router configuration, what is the purpose of setting up QoS (Quality of Service)?
- A. To prioritize network traffic based on specific criteria
 - B. To secure the router from unauthorized access
 - C. To establish trunks between routers
 - D. To exchange routing information with other routers
16. In router configuration, what is the purpose of setting up SSH (Secure Shell) access?
- A. To determine the best path for data transmission
 - B. To prevent unauthorized access to the router
 - C. To exchange routing information with other routers
 - D. To manage VLANs on the router
17. In the TCP/IP model, which layer is responsible for segmenting, reassembling, and error checking of data?
- A. Data Link Layer
 - B. Transport Layer
 - C. Network Layer
 - D. Physical Layer
18. Noah is installing a wireless security camera system outside his home. He needs a band that provides the longest range to ensure all cameras stay connected. Which band should he use?
- A. 2.4 GHz
 - B. 5 GHz
 - C. 11 GHz
 - D. 60 GHz
19. A network administrator configures a route that is used as a backup route when the primary route is unavailable. What type of route is this in routing?
- A. Direct Route
 - B. Static Route
 - C. Default Route
 - D. Floating Route

20. A network administrator configures a route that is used when all other routes fail to reach a specific destination. What type of route is this in routing?
- A. Direct Route
 - B. Static Route
 - C. Default Route
 - D. Last Resort Route
21. A network administrator configures a route with a specific path to a destination network. What type of route is this in routing?
- A. Direct Route
 - B. Static Route
 - C. Default Route
 - D. Dynamic Route
22. A network administrator is configuring a switch to handle multiple VLANs. Which configuration step is necessary to allow the switch to forward traffic between VLANs?
- A. Create a VLAN interface
 - B. Enable trunking on the switch ports
 - C. Implement a VLAN access list
 - D. Configure a VLAN routing protocol
23. A network administrator is configuring FHRP to ensure high availability and load balancing of gateway addresses. Which technology allows for automatic failover and load sharing between routers?
- A. GLBP (Gateway Load Balancing Protocol)
 - B. NAT (Network Address Translation)
 - C. VPN (Virtual Private Network)
 - D. Proxy
24. A network administrator is configuring IPv6 addresses for devices that require a stable address over multiple network reconfigurations. Which addressing method should be used for this requirement?
- A. EUI-64
 - B. SLAAC
 - C. DHCPv6
 - D. Static Address Assignment
25. A network administrator is implementing FHRP to ensure high availability of network resources. Which protocol allows for automatic failover in case of primary router failure?
- A. VRRP (Virtual Router Redundancy Protocol)
 - B. VPN (Virtual Private Network)
 - C. DHCP (Dynamic Host Configuration Protocol)
 - D. DNS (Domain Name System)

26. A network administrator is troubleshooting connectivity issues between IPv6 hosts on the same network segment. Which protocol should be used to verify the link-layer address of a neighboring device?
- A. ARP
 - B. DHCPv6
 - C. ICMPv6
 - D. NDP
27. A network administrator needs to configure VLANs on a switch to improve network performance and security. Which step should be taken to assign ports to specific VLANs?
- A. Create VLANs on the switch
 - B. Enable trunking between switches
 - C. Implement VLAN access lists
 - D. Configure VLAN membership modes
28. A network administrator needs to control traffic flow between VLANs for security reasons. Which network feature should be configured to restrict access between VLANs?
- A. VLAN access list
 - B. VLAN trunking
 - C. VLAN tagging
 - D. VLAN pruning
29. A network administrator wants to create logical groupings of devices within the same broadcast domain. Which feature should be implemented to achieve this?
- A. NAT
 - B. ACL
 - C. VLAN
 - D. OSPF
30. A network administrator wants to divide the network into separate collision domains. Which device is used to achieve this?
- A. Router
 - B. Switch
 - C. Hub
 - D. Repeater
31. A network administrator wants to ensure that all IPv6 traffic within the same subnet is limited to that subnet. Which type of address should be used to achieve this restriction?
- A. Global Unicast
 - B. Link-Local
 - C. Site-Local
 - D. Unique Local

32. A network administrator wants to implement security protocols to encrypt data transmission over the network. Which technology should be used for this purpose?
- A. VPN
 - B. VLAN
 - C. NAT
 - D. ACL
33. A network administrator wants to synchronize VLAN information across multiple switches in the network. Which feature should be enabled for this purpose?
- A. VLAN
 - B. Port Security
 - C. Trunking
 - D. VTP
34. A network where devices are connected in a closed loop, passing data in one direction is an example of which network topology?
- A. Mesh Topology
 - B. Ring Topology
 - C. Star Topology
 - D. Bus Topology
35. A network where devices are connected in a linear fashion with each device connected to two others is an example of which network topology?
- A. Mesh Topology
 - B. Ring Topology
 - C. Star Topology
 - D. Bus Topology
36. The router uses the BGP routing protocol. What type of routing algorithm does BGP use?
- A. Link-State
 - B. Distance-Vector
 - C. Path-Vector
 - D. Hybrid
37. A router uses the RIP routing protocol. What type of routing algorithm does RIP use?
- A. Link-State
 - B. Distance-Vector
 - C. Path-Vector
 - D. Hybrid

38. An organization has multiple departments requiring isolation of their network traffic. Which feature allows for logical segmentation on a single physical switch?
- A. VLAN
 - B. VPN
 - C. Firewall
 - D. Proxy
39. An organization wants to implement a feature that allows devices to verify the uniqueness of their IPv6 addresses before assignment. Which protocol should be used for this purpose?
- A. ARP
 - B. ICMPv6
 - C. DHCPv6
 - D. Duplicate Address Detection
40. An organization wants to implement a feature that provides stable and consistent IPv6 addresses to devices on the network. Which mechanism should be used for this purpose?
- A. SLAAC
 - B. DHCPv6
 - C. Temporary Address Autoconfiguration
 - D. Stateful Address Autoconfiguration
41. In a LAN/WAN design, a network administrator wants to ensure redundancy and load balancing across multiple routers. Which routing protocol should be implemented for this purpose?
- A. EIGRP
 - B. ACL
 - C. NAT
 - D. VLAN
42. In a LAN/WAN design, multiple switches are interconnected to create a redundant network. What technology is used to prevent network loops and ensure a loop-free topology?
- A. OSPF
 - B. STP
 - C. VPN
 - D. VLAN
43. What is the advantage of using IPv6 over IPv4 in terms of security?
- A. Improved encryption protocols
 - B. Enhanced address space for unique addresses
 - C. Faster data transmission
 - D. Stronger firewall capabilities

44. What is the function of the Multicast address in IPv6 networks?
- A. To communicate with multiple hosts on the network
 - B. To identify the network portion of an IP address
 - C. To manage network traffic based on IP addresses
 - D. To prioritize network traffic based on specific criteria
45. What is the purpose of the Global Unicast address in IPv6?
- A. To communicate across multiple subnets
 - B. To secure the router from unauthorized access
 - C. To manage network traffic based on IP addresses
 - D. To prioritize network traffic based on specific criteria
46. What is the role of OSPF (Open Shortest Path First) in routing protocols?
- A. To manage network traffic based on IP addresses
 - B. To determine the best path for data transmission
 - C. To exchange routing information with other routers using link-state advertisements
 - D. To secure the router from unauthorized access
47. When configuring ACLs for IPv4, what is the purpose of using wildcard masks in defining rules?
- A. To determine the best path for data transmission
 - B. To identify the network portion of an IP address
 - C. To manage network traffic based on IP addresses
 - D. To secure the router from unauthorized access
48. When implementing VLANs, what is the significance of a trunk link between switches?
- A. It carries traffic for multiple VLANs
 - B. It encrypts data transmission
 - C. It blocks unauthorized access
 - D. It prioritizes network traffic
49. Which routing protocol is considered a distance vector routing protocol?
- A. OSPF
 - B. RIP
 - C. EIGRP
 - D. BGP
50. Which routing protocol uses a composite metric based on bandwidth, delay, load, and reliability for path selection?
- A. OSPF
 - B. RIP
 - C. EIGRP
 - D. BGP